PRISMA 2020 Checklist

Human Dependence on Large Language Models: A Systematic Review and Taxonomy of Cognitive, Behavioral, Decision-Making, and Emotional Impacts

[Authors anonymized for double-anonymized review]

Table S5. PRISMA 2020 checklist for the main manuscript.

Section and topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title page (p. 1): "... A Systematic Review and Taxonomy ..."
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract (p. 1); see Table S6
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Sections 1.1–1.3 (pp. 1–2); Table 1 (p. 3)
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section 2, RQ1–RQ5 (pp. 2–3); Section 1.4 (p. 2)
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 3.3 and Table 2 (p. 4); grouping for syntheses in Section 3.6 (p. 5)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 3.2 (pp. 3–4); Supplementary S1.3, Table S1. All five databases and citation/update searching last searched 30 September 2026
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 3.2 (pp. 3–4); Supplementary S1.1–S1.2 (full strings, field tags, date limits and platform-specific deviations for all five databases; Figs. S1–S5)
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 3.4 (p. 4); Supplementary S1.4. Deduplication in Rayyan; two-stage screening; AI-proposed preliminary codes, final decisions by the authors; no independent second screener (also Section 6.4, p. 15)
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 3.5 (pp. 4–5); Supplementary S2.1. Extraction performed and checked by the authors; investigators not contacted (Section 6.4, p. 15); AI tool use described in the Declaration of generative AI (p. 16)
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 3.5 (p. 4): dependence dimensions (cognitive, behavioral, decision-making, emotional) and findings relevant to RQ1–RQ5; quantitative results recorded only as reported in abstracts or records; operational definitions in Section 1.2 and Table 4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 3.5 (pp. 4–5): bibliographic details, application domain, research design, participant characteristics (when available); category definitions for domain and design; Supplementary S2.1
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 3.5 (p. 5): MMAT 2018; two reviewers rated all 91 studies independently; agreement 90.33%, Cohen's $\kappa \approx 0.71$; consensus resolution; Supplementary S2.2

Continued on next page

Table 1 (continued)

Section and topic	Item #	Checklist item	Location where item is reported
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not applicable: narrative synthesis; no harmonised effect measure. Results presented descriptively as reported by primary studies (Section 3.5, p. 4)
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Section 3.6 (p. 5): studies mapped to the four dependence dimensions and to thematic clusters by RQ; Table 4 (p. 9)
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Not applicable: no data conversion or imputation; numerical findings recorded as reported (Section 3.5, p. 4)
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Sections 3.5–3.6 (pp. 4–5); Tables 3, 4, 6 and 7; Fig. 2 (evidence map)
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 3.6 (p. 5): convergent narrative synthesis with design-based weighting; rationale for not performing meta-analysis in Section 6.4 (p. 15)
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Section 3.6 (p. 5): moderators and design-dependent contradictions contrasted narratively within each cluster; no statistical subgroup analysis
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not performed; evidence weighted by design and MMAT appraisal (Section 3.5, p. 5)
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not formally assessed; potential publication bias from language and peer-review restrictions acknowledged in Section 6.4 (p. 16)
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	No formal framework (e.g. GRADE, GRADE-CERQual) applied; strength of evidence judged by design hierarchy and MMAT ratings (Section 3.5, p. 5)
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 4.1 (p. 5); Fig. 1, PRISMA 2020 flow diagram (p. 6); Supplementary S1.3–S1.4, Tables S1–S2
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Section 4.1 (p. 5); Fig. 1 (p. 6); Supplementary S1.4, Table S2 (full-text exclusions by reason code). Record-level log available on request
Study characteristics	17	Cite each included study and present its characteristics.	Section 4.2 (p. 5); Table 3 (p. 7); Fig. 2 (p. 8); Table 7 (pp. 21–25); Supplementary S2.1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 4.3 (pp. 6–7); Table 3 (p. 7); Supplementary S2.2, Table S4 (criterion-level consensus ratings for all 91 studies)
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 7 (pp. 21–25): principal findings of each study as reported; Sections 4.4–4.7 (pp. 7–12). Effect estimates and intervals not re-extracted (Section 6.4)
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Sections 4.2–4.3 (pp. 5–7); contributing studies characterised within each cluster in Sections 4.4–4.7 (pp. 7–12); Table 4 (p. 9)
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Not applicable (no statistical synthesis). Narrative synthesis results in Sections 4.4–4.7 (pp. 7–12) and Table 6 (p. 15)

Continued on next page

Table 1 (continued)

Section and topic	Item #	Checklist item	Location where item is reported
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section 4.5, "Moderators" (p. 10); Section 4.7, "Displacement of human relational support" (p. 12); Section 5 (pp. 12–13)
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable (no sensitivity analyses conducted)
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not applicable (not formally assessed); see Section 6.4 (p. 16)
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not applicable (no formal certainty assessment); strength of evidence discussed in Sections 4.3, 4.7 and 5 and Table 5 (pp. 6–13)
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Sections 6.1–6.2 (pp. 13–14); Figs. 3–4 (p. 14)
	23b	Discuss any limitations of the evidence included in the review.	Section 6.4 (pp. 15–16); Section 5 and Table 5 (pp. 12–13)
	23c	Discuss any limitations of the review processes used.	Section 6.4 (pp. 15–16)
	23d	Discuss implications of the results for practice, policy, and future research.	Section 5 and Table 5 (pp. 12–13); Section 6.3 (pp. 14–15); Section 7 (p. 16)
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 3.1 (p. 3): not prospectively registered
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 3.1 (p. 3): no public protocol; research questions, eligibility criteria and coding categories specified a priori
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not applicable (not registered); platform-specific search deviations reported in Section 3.2 (p. 4) and Supplementary S1.1
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Funding statement (p. 16) [to be added before submission]
Competing interests	26	Declare any competing interests of review authors.	Declaration of competing interest (p. 16)
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Data availability statement (p. 16); Supplementary S1 (search strings, audit trail, screening protocol) and S2 (extraction sheet, `extraction_sheet.xlsx`, MMAT ratings). Screening log on request; no analytic code (narrative synthesis)

Table S6. PRISMA 2020 for Abstracts checklist.

Section and topic	Item #	Checklist item	Reported	Location / note
TITLE				
Title	1	Identify the report as a systematic review.	Yes	Title
BACKGROUND				
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes	Abstract, sentences 1–2 and taxonomy aim
METHODS				
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes	Peer-reviewed empirical studies, 1 December 2022 to 30 September 2026
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes	Five named databases plus citation and update searching; searched 30 September 2026
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes	MMAT 2018, two independent reviewers
Synthesis of results	6	Specify the methods used to present and synthesise results.	Partly	Taxonomy-based synthesis described; consider naming "convergent narrative synthesis"
RESULTS				
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes	91 studies; domains and designs summarised (participant totals not pooled)
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes	Main findings by dimension; direction of effects stated (narrative)
DISCUSSION				
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes	Unstated assessor blinding, nonresponse bias, few longitudinal and clinical studies
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes	Framework, reliance calibration matrix, research agenda and implications
OTHER				
Funding	11	Specify the primary source of funding for the review.	No	Not stated in abstract (journal abstract format); see item 25 of Table S5
Registration	12	Provide the register name and registration number.	No	Not registered (Section 3.1); optionally state in abstract

Notes. Page, section, table, and figure numbers refer to the main manuscript as submitted; "Supplementary S1–S3" and Tables S1–S6 and Figs. S1–S5 refer to this supplementary document. Checklist items are reproduced from Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;372:n71. doi:10.1136/bmj.n71 (CC BY 4.0). Items marked in red require attention before submission.